

Problem-Solution Fit

Date	27 June 2026
Team ID	xxxxxx
Project Name	AI based crop recommendation
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Many farmers struggle to select the most suitable crop because crop growth depends on soil nutrients and environmental conditions. Incorrect crop selection can result in low productivity, financial losses, and inefficient use of agricultural resources. OptiCrop helps farmers make better cultivation decisions by using Machine Learning to analyze soil and climate parameters and recommend the most suitable crop.

1. CUSTOMER SEGMENT(S) [CS] • Farmers • Agricultural Students • Agricultural Extension Officers • Researchers in Precision Agriculture	5. AVAILABLE SOLUTIONS Pros: Traditional farming experience, agricultural experts, soil testing methods. Cons: Time-consuming, requires expert knowledge, does not provide instant crop recommendations.	8. CHANNELS OF BEHAVIOR [CH] *Local agricultural extension centers and government agriculture officers *Agricultural experts and consultants *Farmer communities and peer recommendations
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Difficulty in selecting suitable crops (High) • Lack of knowledge about soil and crop relationships (High) • Wrong crop selection leading to low yield (Medium)	6. CUSTOMER LIMITATIONS • Limited knowledge of advanced agricultural technologies • Limited access to expert agricultural guidance • Budget constraints for frequent soil analysis	9. PROBLEM ROOT / CAUSE [RC] *Lack of personalized crop recommendations based on specific soil conditions *Dependence on traditional farming practices instead of data-driven decisions *Limited awareness and accessibility of precision agriculture technologies

3. TRIGGERS TO ACT [TR] • Preparing a new field for cultivation • Need for higher crop productivity • Requirement for quick crop selection based on soil conditions	7. BEHAVIOR + ITS INTENSITY [BE] • Frequently seek advice from agricultural experts. • Depend on traditional farming methods. • Require quick and reliable crop selection support.	10. YOUR SOLUTION [SL] OptiCrop uses Machine Learning to analyze soil nutrients (N, P, K) and environmental parameters such as temperature, humidity, pH, and rainfall.
4. EMOTIONS BEFORE / AFTER [EM] Before: Confused, uncertain, worried about crop failure. After: Confident, informed, and able to make better farming decisions.		